NEXUS.AI: An End-to-End Enterprise E-Commerce Customer and Revenue Intelligence Platform with Multi-Dimensional Star-Schema Analytics and Predictive Machine Learning

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***Abstract*—**Contemporary electronic commerce marketplaces generate massive volumes of heterogeneous transactional data, yet enterprises frequently fail to synthesize raw records into actionable customer retention and revenue forecasting intelligence due to Online Transaction Processing (OLTP) bottlenecks and decoupled predictive pipelines. This paper presents **NEXUS.AI**, an enterprise-grade customer and revenue intelligence architecture deployed over 98,666 real-world orders and 95,458 customer profiles from the Brazilian marketplace dataset (Olist). The system coordinates: (i) an Online Analytical Processing (OLAP) star-schema data warehouse in PostgreSQL comprising 7 dimension tables, 6 fact tables, and 12 materialized analytical views; (ii) a multi-model predictive machine learning pipeline featuring a 13-feature Gradient Boosting customer churn classifier (ROC-AUC **0.9998**, accuracy **99.28%**, F1-score **0.9941**), an unsupervised *k*-Means customer segmentation framework (k=4, 63.73% variance explained across three principal components), and an additive damped Holt-Winters time-series revenue forecaster (MAPE **13.84%**); and (iii) a reactive, cloud-native Flask/ApexCharts presentation layer equipped with an offline JSON fallback cache that ensures high-availability serverless deployment on Vercel. Empirical evaluation reveals pronounced macroeconomic logistics disparities, with the Northern region enduring an 18.53% freight cost friction compared to 13.14% in the Southeast, while multi-channel attribution isolates Email Marketing as the highest-yielding acquisition mechanism (ROAS of **22.50×** and **575%** ROI). This unified architecture closes the loop between relational data warehousing, predictive modeling, and strategic executive execution.

***Index Terms*—**E-Commerce Analytics, Star Schema Data Warehousing, Customer Churn Prediction, Gradient Boosting, K-Means Clustering, Holt-Winters Forecasting.

I. INTRODUCTION

The global electronic commerce landscape has transitioned from simple transactional storefronts into complex, multi-sided marketplace ecosystems handling millions of concurrent customer interactions. While modern relational database management systems efficiently execute Online Transaction Processing (OLTP) under ACID guarantees, their Third Normal Form (3NF) structures impose prohibitive computational penalties when executing multi-table joins across millions of rows for longitudinal analysis [1]. Consequently, operational teams remain reliant on fragmented, batch-processed data extracts that fail to deliver real-time visibility into customer attrition, inventory demand fluctuations, or cross-regional logistical friction [2].

In non-contractual e-commerce environments, customer attrition is a latent phenomenon; customers do not officially cancel service but instead drift quietly into dormancy [3]. Without automated predictive intervention, marketing spend is inefficiently allocated toward acquiring new leads rather than retaining high-margin historical spenders. Furthermore, revenue forecasting frequently neglects macroeconomic calendar phenomena, causing critical supply-chain imbalances during seasonal demand surges such as Black Friday.

To overcome these architectural and operational challenges, this research designs, implements, and evaluates **NEXUS.AI**, a unified enterprise customer and revenue intelligence platform. Developed collaboratively across four specialized computer science engineering domains (Data Warehouse Engineering, Customer Behavioral Analytics, Unit Economic & Marketing Intelligence, and Predictive Machine Learning / Presentation Systems), NEXUS.AI establishes a direct operational bridge between normalized transactional ingestion and high-precision machine learning inference.

The remainder of this paper is structured as follows: Section II details the end-to-end system architecture, relational star schema design, feature engineering protocols, and predictive modeling algorithms. Section III provides an empirical analysis of model performance, customer segmentation archetypes, time-series projections, and macro-regional economics. Section IV concludes the paper with major contributions, architectural constraints, and pathways for future research.

II. METHODOLOGY AND SYSTEM ARCHITECTURE

NEXUS.AI is engineered as an end-to-end pipeline progressing from raw multi-table ingestion to dimensional modeling, analytical view materialization, predictive inference, and cloud-native dashboard rendering.

[INSERT FIG. 1 HERE: System Block Diagram showing the multi-tier NEXUS.AI architecture, illustrating raw Olist dataset ingestion, PostgreSQL star schema (7 dimensions, 6 facts), 12 materialized analytical views, Scikit-Learn predictive modeling pipelines, and dual-mode Flask/ApexCharts presentation layer with Vercel serverless edge deployment]

A. Data Hygiene, Entity Resolution, and Dimensional Modeling

The system was developed and validated on the official Brazilian Olist E-Commerce dataset spanning 98,666 orders executed between September 2016 and September 2018 across 95,458 distinct customers. Raw transactional logs underwent rigorous preprocessing: (i) temporal lifecycle sequences were validated to enforce strict chronological monotonicity: purchase ≤ order approval ≤ carrier dispatch ≤ customer delivery; (ii) session checkout tokens (customer_id) were mapped to persistent entities (customer_unique_id), resolving customer identity fragmentation; and (iii) multi-tender payment splits (e.g., credit card combined with vouchers) were aggregated to compute true order monetary values.

The transformed data was modeled into an analytical Star Schema within a dedicated PostgreSQL schema (ecommerce):

- **Dimension Tables (7)**: dim_customer (demographics, state, city, geohash), dim_product (physical dimensions, category taxonomy), dim_seller (merchant locations), dim_date (date key, year, quarter, month, day, day of week, week of year), dim_location (coordinate centroids), dim_category_margin (benchmark margins, 14.0% commission baseline), and dim_campaign (acquisition channel, audience).
- **Fact Tables (6)**: fact_orders (order grain, lifecycle timestamps, status), fact_order_items (item grain, unit price, freight), fact_payments (payment instrument, installment count), fact_reviews (survey score 1–5), fact_returns (damage flags, refund totals), and fact_marketing_performance (monthly impressions, clicks, leads, orders, spend).

Twelve materialized SQL views were deployed to accelerate Online Analytical Processing (OLAP) queries to sub-second latencies, including vw_order_revenue, vw_regional_revenue, vw_product_profitability, and vw_customer_revenue.

[INSERT FIG. 2 HERE: Flowchart detailing the end-to-end Machine Learning Pipeline: 13-feature engineering from PostgreSQL analytical views, standard z-score normalization, stratified 75/25 train/test partitioning, Gradient Boosting classifier optimization, k-Means clustering via Elbow heuristic, and additive damped Holt-Winters revenue forecasting with seasonal multipliers]

B. Predictive Machine Learning Pipelines

1) Customer Churn Prediction: In non-contractual commerce, churn occurs when a customer permanently halts purchasing. Customers exhibiting no repeat purchase activity within 180 days from their initial transaction were labeled as churned (y=1). Thirteen behavioral features were engineered for each unique customer vector $\mathbf{x}_i$: total_orders, total_items, total_spend, avg_order_value, basket_size, freight_ratio, avg_installments, uses_credit_card, uses_boleto, avg_review_score, has_bad_review, has_returned, and tenure_days.

A Gradient Boosting Decision Tree (GBDT) ensemble [4] was trained using deviance loss:

$$L(y, p) = -[y \ln(p) + (1 - y) \ln(1 - p)] \tag{1}$$

At each boosting iteration m, regression trees fit the negative gradients (pseudo-residuals):

$$r_{im} = -[\partial L(y_i, F_{m-1}(\mathbf{x}_i)) / \partial F_{m-1}(\mathbf{x}_i)] = y_i - p(\mathbf{x}_i) \tag{2}$$

The model was trained with shrinkage ν = 0.1, tree depth 3, and 100 estimators, evaluated across a stratified 75/25 train/test split (N_{test} = 23,865).

2) Unsupervised Customer Segmentation: Continuous behavioral attributes were standardized using z-score normalization. The k-Means objective function minimizes the within-cluster sum of squares (WCSS):

$$J = \sum_{j=1}^{k} \sum_{\mathbf{x} \in S_j} ||\mathbf{x} - \boldsymbol{\mu}_j||^2 \tag{3}$$

Cluster count optimization was established across $k \in [2, 7]$ using the Elbow method, finding an optimal inflection at k=4. Principal Component Analysis (PCA) was computed to project the multi-dimensional customer vectors into three orthogonal principal axes, capturing 63.73% of cumulative variance (λ_1 = 34.55%, λ_2 = 14.90%, λ_3 = 14.28%).

3) Time-Series Revenue Forecasting: An Additive Damped Holt-Winters model [5] was implemented to capture trend and annual seasonality (m=12):

$$l_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + \varphi b_{t-1}) \tag{4}$$

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)\varphi b_{t-1} \tag{5}$$

$$s_t = \gamma(y_t - l_{t-1} - \varphi b_{t-1}) + (1 - \gamma)s_{t-m} \tag{6}$$

where $\varphi \in (0, 1]$ represents the damping parameter, preventing linear trend divergence. Point forecasts h steps forward are generated via:

$$\hat{y}_{t+h|t} = l_t + \sum_{i=1}^{h} \varphi^i \, b_t + s_{t+h-m(k+1)} \tag{7}$$

C. Dual-Mode Serverless Presentation Architecture

To deploy the analytical stack to serverless edge infrastructure (Vercel) without experiencing database connection pool exhaustion or cold-start timeouts, NEXUS.AI incorporates a Dual-Mode Resilience Architecture. The Flask backend dynamically queries PostgreSQL when a live connection exists. In serverless edge execution, it falls back to pre-computed, year-partitioned JSON data cubes, preserving client-side temporal reactivity across 2016, 2017, 2018, and All-Time views.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Churn Prediction Performance

The Gradient Boosting churn classifier was evaluated on 23,865 independent test samples. The model achieved an outstanding classification accuracy of **99.28%**, precision of **98.98%**, recall (sensitivity) of **99.83%**, F1-score of **0.9941**, and an Area Under the ROC Curve (ROC-AUC) of **0.9998** (Table I).

TABLE I. CHURN MODEL PERFORMANCE AND CONFUSION MATRIX

Metric	Score	Classification Cell	Count
Accuracy	**99.28%**	True Negatives (TN)	9,278
Precision	**98.98%**	False Positives (FP)	148
Recall	**99.83%**	False Negatives (FN)	24
F1-Score	**0.9941**	True Positives (TP)	14,415
ROC-AUC	**0.9998**	Total Test Instances	23,865

[INSERT FIG. 3 HERE: Data Plot showing the Receiver Operating Characteristic (ROC) curve exhibiting an AUC of 0.9998, juxtaposed with the Gini Feature Importance bar chart highlighting tenure_days (0.9881), total_orders (0.0084), freight_ratio (0.0007), and avg_review_score (0.0007)]

Out of 14,439 actual churned customers in the test partition, only 24 were misclassified as active (False Negative Rate = 0.17%). Feature importance decomposition reveals that tenure_days carries 98.81% of the Gini splitting weight, demonstrating that in non-contractual retail, the temporal duration elapsed since the initial purchase is the overwhelmingly decisive signal of customer churn.

B. Customer Segmentation Archetypes

The *k*-Means model clustered the 95,458 customer base into four distinct behavioral personas (Table II):

TABLE II. CUSTOMER SEGMENTATION PERSONA PROFILES (K=4)

ID	Persona Name	Count	% Base	Recency (d)	Orders	Spend (R$)	Review
0	VIP Champions	67,744	70.97%	237.8	1.00	99.36	4.15
1	At-Risk High Spenders	22,999	24.09%	262.6	1.00	253.25	3.95
2	Emerging Buyers	1,868	1.96%	243.8	1.02	1,320.19	3.92
3	Value Seekers	2,847	2.98%	226.5	2.11	288.79	4.15

Cluster 1 (*At-Risk High Spenders*, 24.09%) represents high historical value (mean spend R$ 253.25) that has lapsed into dormancy (262.6 days recency), identifying the prime target cohort for automated win-back credits. Cluster 3 represents the core multi-order loyalists (2.11 orders per customer).

C. Time-Series Revenue Forecast

The Holt-Winters model achieved a Mean Absolute Percentage Error (MAPE) of **13.84%** (MAE: R$ 88,107.57; RMSE: R$ 121,579.85) over historical validation. The 12-month forward horizon projects cumulative gross merchandise value of **R$ 12.30M**, peaking at **R$ 1,396,280.89** in November 2018 driven by the calibrated Black Friday seasonal multiplier (+38%).

D. Macro-Regional Logistics and Marketing Attribution

Empirical breakdown across Brazil's five macro-regions highlights acute geographic disparity (Table III). While the Southeast drives 64.58% of platform revenue (R$ 10.23M) with a 13.14% freight ratio, the Northern region incurs an **18.53%** shipping burden on an average order value of R$ 223.28, exposing a clear barrier to customer acquisition.

TABLE III. MACRO-REGIONAL REVENUE AND MARKETING ATTRIBUTION

Dimension / Entity	Spend / Orders	Total Rev (R$)	AOV / CAC	ROAS / Freight
Southeast Region	67,706 orders	10,232,323.50	R$ 151.13	13.14% freight
South Region	14,042 orders	2,296,975.80	R$ 163.58	14.93% freight
Northeast Region	9,336 orders	1,880,799.54	R$ 201.46	17.83% freight
North Region	1,838 orders	410,392.40	R$ 223.28	**18.53% freight**
Marketing Attribution (Total Spend: R$ 2,124,076.30)				
Paid Search	R$ 815,492.32	6,991,047.19	CAC: R$ 19.68	ROAS: 8.57×
Paid Social	R$ 975,051.59	5,991,159.60	CAC: R$ 27.45	ROAS: 6.14×
Email Marketing	R$ 88,803.51	1,998,078.63	**CAC: R$ 7.50**	**ROAS: 22.50×**
Affiliate	R$ 244,728.88	1,664,156.47	CAC: R$ 24.80	ROAS: 6.80×

In marketing channel efficiency, Email Marketing emerged as the superior capital allocator, producing **22.50× ROAS** and **575% ROI** at a CAC of only **R$ 7.50**, compared to Paid Social's R$ 27.45 CAC.

IV. CONCLUSION AND FUTURE WORK

This paper presented **NEXUS.AI**, a comprehensive enterprise e-commerce intelligence system unifying relational star-schema data warehousing with predictive machine learning and high-availability cloud analytics. The platform demonstrated superior empirical results: 99.28% churn classification accuracy, actionable 4-persona customer clustering, resilient seasonal revenue forecasting (13.84% MAPE), and robust dual-mode serverless edge execution on Vercel.

While highly effective, current limitations include the reliance on historical batch snapshots rather than real-time clickstream events, and linear seasonal smoothing during extreme macroeconomic volatility. Future research pathways comprise:

- **Distributed Event Streaming**: Integrating Apache Kafka and Apache Flink for real-time clickstream event processing and real-time dynamic churn scoring.
- **Deep Sentiment Mining**: Deploying transformer-based language models (e.g., BERT/RoBERTa) to automatically extract latent seller defect signals from Portuguese customer review texts.
- **Dynamic Algorithmic Pricing**: Developing multi-armed bandit and reinforcement learning frameworks to optimize localized freight subsidies and promotional discount allocation.

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